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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 20	
TITLE: JDBC INSERT, UPDATE, and DELETE Operations	

Problem Statement

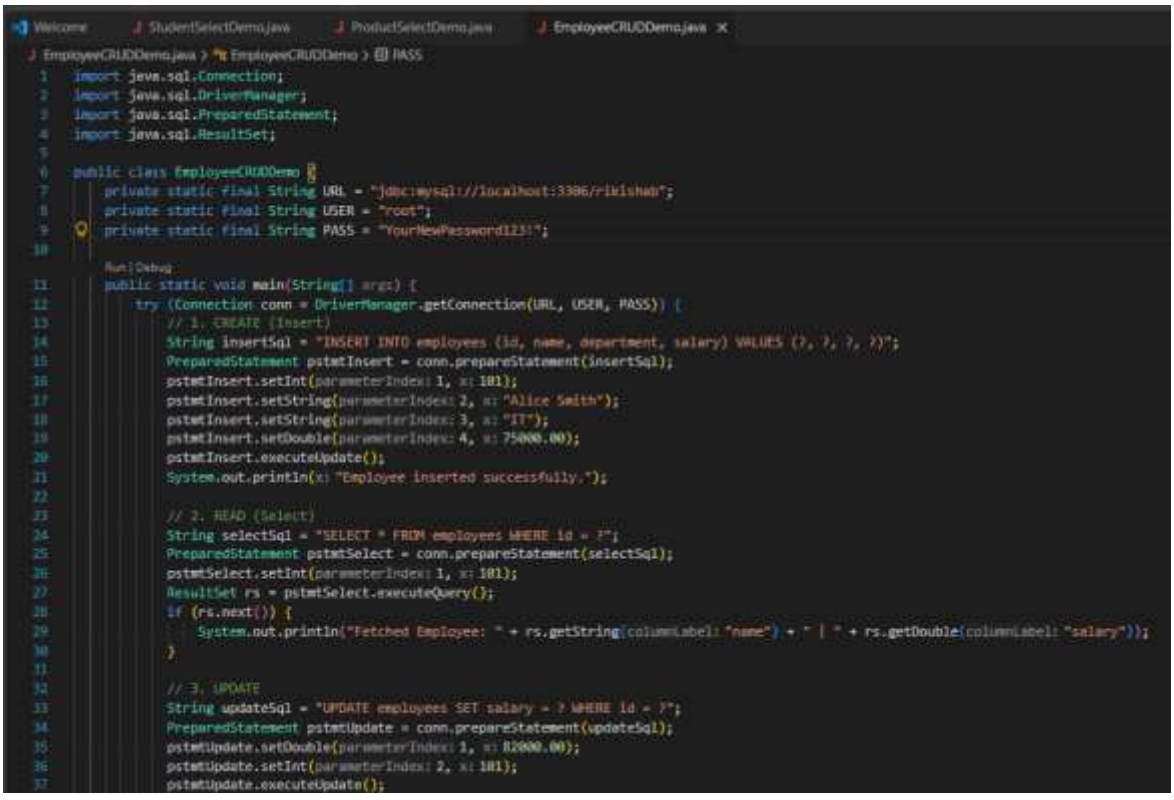
Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives

To perform database modification operations using JDBC.

1. **Create a Java application to manage employee records using CRUD operations**

Programme



```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

public class EmployeeCRUDDemo {
    private static final String URL = "jdbc:mysql://localhost:3306/rikishadb";
    private static final String USER = "root";
    private static final String PASS = "YourNewPassword123!";

    public static void main(String[] args) {
        try (Connection conn = DriverManager.getConnection(URL, USER, PASS)) {
            // 1. CREATE (Insert)
            String insertSql = "INSERT INTO employees (id, name, department, salary) VALUES (?, ?, ?, ?)";
            PreparedStatement pstmtInsert = conn.prepareStatement(insertSql);
            pstmtInsert.setInt(parameterIndex: 1, x: 101);
            pstmtInsert.setString(parameterIndex: 2, x: "Alice Smith");
            pstmtInsert.setString(parameterIndex: 3, x: "IT");
            pstmtInsert.setDouble(parameterIndex: 4, x: 75000.00);
            pstmtInsert.executeUpdate();
            System.out.println(x: "Employee inserted successfully.");

            // 2. READ (Select)
            String selectSql = "SELECT * FROM employees WHERE id = ?";
            PreparedStatement pstmtSelect = conn.prepareStatement(selectSql);
            pstmtSelect.setInt(parameterIndex: 1, x: 101);
            ResultSet rs = pstmtSelect.executeQuery();
            if (rs.next()) {
                System.out.println("Fetched Employee: " + rs.getString(columnLabel: "name") + " | " + rs.getDouble(columnLabel: "salary"));
            }

            // 3. UPDATE
            String updateSql = "UPDATE employees SET salary = ? WHERE id = ?";
            PreparedStatement pstmtUpdate = conn.prepareStatement(updateSql);
            pstmtUpdate.setDouble(parameterIndex: 1, x: 82000.00);
            pstmtUpdate.setInt(parameterIndex: 2, x: 101);
            pstmtUpdate.executeUpdate();
```

```
            System.out.println(x: "Employee salary updated.");

            // 4. DELETE
            String deleteSql = "DELETE FROM employees WHERE id = ?";
            PreparedStatement pstmtDelete = conn.prepareStatement(deleteSql);
            pstmtDelete.setInt(parameterIndex: 1, x: 101);
            pstmtDelete.executeUpdate();
            System.out.println(x: "Employee record deleted.");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output

```
Problems 11   Output   Debug Console   Terminal   Ports

PS C:\Users\RIKISHA\PROGRAMMING\Java\src>  & 'C:\Program Files\Java\jdk-26.0.1
oyeeCRUDDemo'
Employee inserted successfully.
Fetched Employee: Alice Smith | 75000.0
Employee salary updated.
Employee record deleted.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. Create a Java application to manage student records using CRUD operations. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

Programme

```
Welcome   StudentSelectDemo.java   ProductSelectDemo.java   EmployeeCRUDDemo.java   StudentCRUDDemo.java
StudentCRUDDemo.java > StudentCRUDDemo
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

public class StudentCRUDDemo {
    private static final String URL = "jdbc:mysql://localhost:3306/rikishab";
    private static final String USER = "root";
    private static final String PASS = "YourNewPassword123";

    Run | Debug
    public static void main(String[] args) {
        try (Connection conn = DriverManager.getConnection(URL, USER, PASS)) {
            // CREATE
            String insertSql = "INSERT INTO students (roll_no, name, course, marks) VALUES (?, ?, ?, ?)";
            try (PreparedStatement pstmt = conn.prepareStatement(insertSql)) {
                pstmt.setInt(parameterIndex: 1, x: 1001);
                pstmt.setString(parameterIndex: 2, x: "Rikisha Bhattacharya");
                pstmt.setString(parameterIndex: 3, x: "Computer Science");
                pstmt.setDouble(parameterIndex: 4, x: 92.5);
                pstmt.executeUpdate();
                System.out.println(x: "Student record added.");
            }

            // READ
            String selectSql = "SELECT * FROM students WHERE roll_no = ?";
            try (PreparedStatement pstmt = conn.prepareStatement(selectSql)) {
                pstmt.setInt(parameterIndex: 1, x: 1001);
                ResultSet rs = pstmt.executeQuery();
                while (rs.next()) {
                    System.out.println("Student: " + rs.getString(columnLabel: "name") + " - Marks: " + rs.getDouble(columnLabel: "marks"));
                }
            }

            // UPDATE
            String updateSql = "UPDATE students SET marks = ? WHERE roll_no = ?";
            try (PreparedStatement pstmt = conn.prepareStatement(updateSql)) {
                pstmt.setDouble(parameterIndex: 1, x: 95.0);
                pstmt.setInt(parameterIndex: 2, x: 1001);
```

```
            pstmt.executeUpdate();
            System.out.println(x: "Student marks updated.");
        }

        // DELETE
        String deleteSql = "DELETE FROM students WHERE roll_no = ?";
        try (PreparedStatement pstmt = conn.prepareStatement(deleteSql)) {
            pstmt.setInt(parameterIndex: 1, x: 1001);
            pstmt.executeUpdate();
            System.out.println(x: "Student record deleted.");
        }

    } catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

Output

```
Problems 11   Output   Debug Console   Terminal   Ports

PS C:\Users\RIKISHA\PROGRAMMING\Java\src> & 'C:\Program Files\Java\jdk-26.0.
entCRUDDemo'
Student record added.
Student: Rikisha Bhattacharya - Marks: 92.5
Student marks updated.
Student record deleted.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

Conclusion:

Thus, JDBC was successfully used to perform CRUD operations (Create, Read, Update, and Delete) on employee and student records.